

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
 December 2024

MA2002-1- NUMERICAL METHODS
 (For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- While using Newton's backward difference interpolation formula, for any x the value of p is _____
 A) $\frac{x+x_0}{h}$ B) $\frac{x+x_n}{h}$
 C) $\frac{x-x_0}{h}$ D) $\frac{x-x_n}{h}$
- If $f(x) = 2x + 2$, $h = 1$ then $E^{-1}(f(x)) =$ _____
 A) $2x - 2$ B) $2x$
 C) $2x + 2$ D) $2x + 4$
- Expansion of $\Delta^3 y_0$ in terms of entries _____
 A) $y_3 - 3y_2 + 3y_1 + y_0$ B) $y_3 + 3y_2 + 3y_1 + y_0$
 C) $y_3 - 3y_2 + 3y_1 - y_0$ D) $y_4 - 3y_3 + 3y_2 - y_1$
- The n^{th} order forward difference of n^{th} degree polynomial is _____
 A) a constant B) zero
 C) a polynomial of degree n D) a polynomial of degree $n - 1$
- Given

x	0	1	2	3
y	1	3	11	31

 the value of $y'(0) =$ _____
 A) 7 B) 1
 C) 3.5 D) 0.5
- To apply Simpson's $3/8^{\text{th}}$ rule to get an approximate value of $\int_{x_0}^{x_n} f(x)dx$, the number of sub intervals must be a
 A) multiple of 2 B) multiple of 5
 C) multiple of 3 D) multiple of 4
- If $x_0, x_1, x_2, \dots, x_n$ are equally spaced then expression for $\frac{dy}{dx}$ at $x = x_0$ using forward difference operator is _____
 A) $\frac{1}{h}\{\Delta y_0 + \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 + \frac{1}{4}\Delta^4 y_0 + \dots\}$ B) $\frac{1}{h^2}\{\Delta^2 y_0 + \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 + \frac{5}{6}\Delta^5 y_0 + \dots\}$
 C) $\frac{1}{h}\{\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \dots\}$ D) $\frac{1}{h^2}\{\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \dots\}$
- The value of $\int_1^{2.1} \frac{1}{x} dx =$ _____ calculated by using Trapezoidal rule with $h=0.5$.
 A) 0.6851 B) 0.6931
 C) 0.7233 D) 0.7083

9. For the initial value problem $\frac{dy}{dx} = f(x, y)$, $y(x_0) = y_0$, the Taylor's series expansion up to fourth order is _____
- A) $y_0 + h^2 y_0' + \frac{h^3}{2!} y_0'' + \frac{h^4}{3!} y_0''' + \frac{h^5}{4!} y_0^{iv} + \dots$ B) $y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{iv} + \dots$
- C) $y_0 + h y_0' - \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' - \frac{h^4}{4!} y_0^{iv} + \dots$ D) $h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{iv} + \dots$
10. In modified Euler's method, the initial approximation $y_1^{(0)}$ for the initial value problem $\frac{dy}{dx} = x - y^2$, $y(0) = 1$, $h = 0.1$ is _____
- A) 0.8 B) 0.88
C) 0.9 D) 0.99
11. In Runge-Kutta method of order four, the value of k_1 for the initial value problem $\frac{dy}{dx} = x + y$, $y(0) = 1$, $h = 0.2$ is _____
- A) 0.2 B) 0.22
C) 0.1 D) 0.15
12. Milne's predictor formula to find y_4 is _____
- A) $y_4 = y_0 + \frac{4h}{3} (2f_1 + f_2 + 2f_3)$ B) $y_4 = y_2 + \frac{h}{3} (f_2 + 4f_3 + f_4^p)$
C) $y_4 = y_2 + \frac{h}{3} (f_2 - 4f_3 + f_4^p)$ D) $y_4 = y_0 + \frac{4h}{3} (2f_1 - f_2 + 2f_3)$
13. The standard five point formula for solving Laplace's equation $u_{xx} + u_{yy} = 0$ is _____
- A) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ B) $u_{i,j} = u_0 + \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
C) $u_{i,j} = \frac{1}{3} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$ D) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}] - u_{0,0}$
14. For the heat equation $u_t = 0.5u_{xx}$, $h = 1$, $k = 1$, the value of mesh ratio parameter α is _____
- A) 1 B) $\frac{1}{2}$
C) $\frac{1}{4}$ D) 2
15. The equation $u_{tt} = 16u_{xx}$ is classified as _____
- A) Elliptic B) Parabolic
C) Laplace D) Hyperbolic
16. For the wave equation $u_{tt} = u_{xx}$, $h = k = 0.2$, the value of α is _____
- A) 1 B) 1.5
C) 0.5 D) 1.25
17. If $f(x)$ is continuous in some interval $[a, b]$ such that $f(a)f(b) < 0$, then the equation $f(x) = 0$ has _____ root in $[a, b]$.
- A) exactly one B) at least one
C) even number D) odd number
18. The first iterative value of the root of $x^3 - 3x - 5 = 0$ near $x_0 = 3$ using Newton-Raphson method is _____
- A) 2.8888 B) 2.4583
C) 2.2943 D) 2.2791
19. The method which approximates curve as tangent to find root of an algebraic equation is _____
- A) Euler's method B) Modified Euler's method
C) Newton-Raphson method D) Regula-Falsi method
20. The first approximation to the root of $x^3 - 4x - 9 = 0$ in $(2, 3)$ using bisection method is _____
- A) 2.75 B) 2.55
C) 2.5 D) 2.25

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I****Marks BT* CO* PO***

1. a) Using Lagrange's interpolation formula, find the value of y when $x = 10$ from the given data

x	5	6	9	11
y	12	13	14	16

6 L*2 1 1

- b) The velocity v of a particle at distance s from a point on its path is given by the table below

$s(\text{meter})$	0	10	20	30	40	50	60
$v(\text{m/sec})$	47	58	64	65	61	52	38

Estimate the time taken to travel 60m by using Simpson's $1/3^{\text{rd}}$ rule.

6 L2 2 1

- c) Using Newtons interpolation formula find $f(4)$ for the data given below

x	0	1	2	3
$f(x)$	1	2	1	10

4 L1 1 1

2. a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.2$ given

x	1.0	1.2	1.4	1.6
y	2.72	3.32	4.06	4.96

6 L2 2 1

- b) Using Newton's divided difference formula, find the value of y when $x = 1$ from the given data

x	-4	-1	0	2	5
y	1245	33	5	9	1335

6 L2 1 1

- c) Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using Trapezoidal rule taking $h = 1$.

4 L1 2 1

3. a) Derive the Newton's forward difference interpolation formula.

6 L1 1 2

- b) Evaluate $\int_0^1 \frac{x}{1+x^2} dx$, using Simpson's $3/8^{\text{th}}$ rule by considering seven ordinates and hence find approximate value of $\log_e 2$.

6 L2 2 1

- c) Find $\frac{dy}{dx}$ at $x = 1$ given

x	1.0	1.4	1.8	2.2
y	0.7854	0.9505	1.0637	1.1442

4 L1 2 1

Unit – II

4. a) Use Milne's predictor-corrector method to find y at $x = 0.4$ given $\frac{dy}{dx} = 1 + xy^2$, $y(0) = 1$, $y(0.1) = 1.105$, $y(0.2) = 1.223$, $y(0.3) = 1.355$.

6 L2 3 1

- b) Solve the boundary value problem $u_t = u_{xx}$ under the conditions $u(x, 0) = \sin \pi x$ ($0 \leq x \leq 1$), $u(0, t) = u(1, t) = 0$ with $h = 0.2$, $k = 0.02$ for two-time levels.

5 L2 4 1

- c) Using Modified Euler's method, find an approximate value of y when $x = 0.1$, taking $h = 0.1$ for $\frac{dy}{dx} = -xy^2$, $y(0) = 2$. Carry out two iterations in each step.

5 L1 3 1

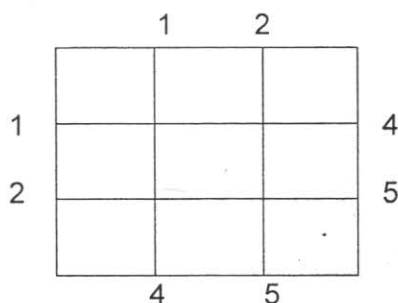
5. a) Solve $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$ with the boundary condition $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \frac{1}{2}x(1-x)$ and $u_t(x, 0) = 0$. Take $h = k = 0.2$. Find the values up to $t=0.6$.

6 L2 4 1

- b) Using fourth order Runge-Kutta method find the value of y at $x = 0.1$ given that $\frac{dy}{dx} = 3x + \frac{y}{2}$, $y(0) = 1$ and $h = 0.1$.

5 L1 3 1

- c) Solve $u_{xx} + u_{yy} = 0$ for the square mesh with the boundary values as shown below:



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|--|---|----|---|---|
| 6. a) Solve $\nabla^2 u = -81xy$, $0 < x, y < 1$, $h = k = 1/3$,
$u(x, 0) = u(0, y) = 0$, $u(x, 1) = 100 = u(1, y)$. | 5 | L2 | 4 | 1 |
| b) Use Adams-Bashforth method to find y at $x = 0.4$ given
$\frac{dy}{dx} = x^2 - y$, $y(0) = 1$, $y(0.1) = 0.90516$, $y(0.2) = 0.82127$,
$y(0.3) = 0.74918$. | 6 | L2 | 4 | 1 |
| c) Using Taylor's series method of order four compute y at $x = 0.1$
given $\frac{dy}{dx} = x + y$, $y(0) = 1$, take $h = 0.1$. | 5 | L | 3 | 1 |

Unit – III

- | | | | | |
|---|---|----|---|---|
| 7. a) Using Gauss Seidel method obtain the solution of the linear system of equations taking the initial approximation to the solution as $x^{(0)} = 0, y^{(0)} = 0, z^{(0)} = 0$. Carry out three iterations.
$27x + 6y - z = 85$
$6x + 15y + 2z = 72$
$x + y + 54z = 110$ | 6 | L1 | 5 | 1 |
| b) Using Newton Raphson's method, obtain the root of the equation $x^3 - 3x + 1 = 0$ near $x = 2$. Carry out three iterations. | 6 | L2 | 5 | 1 |
| c) Using bisection method obtain the root of $x^3 - 9x + 1 = 0$ in $(2, 3)$. Carry out 3 iterations. | 4 | L1 | 5 | 1 |
| 8. a) Using the Regula falsi method find a root of the equation $x^3 - 2x - 5 = 0$ in $(2, 3)$. Carry out three iterations. | 6 | L2 | 5 | 1 |
| b) Using Newton Raphson method solve the following system of nonlinear equations
$x^2 - y^2 = 4$, $x^2 + y^2 = 16$ with $x_0 = y_0 = 2.8$. Find x_1 and y_1 . | 6 | L2 | 5 | 1 |
| c) Using Ordinary iteration method obtain the root of $\cos x = 3x - 1$ near $x = 0$. Carry out three iterations. | 4 | L1 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
